Fourier Series Representation of Periodic Signals

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Motivation

- **Desirable properties for synthesis and analysis**:
  - Set of basic signals (done earlier in a way!)
  - Not just **input signals**, but also **output signals** to be **simple**.
- Recall LSI (LTI) system process:

$$y(t) \quad = x(t) * h(t) = \quad \int_{-\infty}^{\infty} x(u) \cdot \boldsymbol{h(t-u)} du.$$

$$y[n] \quad = x[n] * h[n] \quad \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]}.$$

Motivation

- **Continuous-time**:

$$y(t) \quad = x(t) * h(t) = \quad \int_{-\infty}^{\infty} x(u) \cdot \boldsymbol{h(t-u)} du.$$

- Set input $x(t) = e^{st}$ (complex exponential). We have output as

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} e^{st} \cdot \boldsymbol{h(t-u)} du \\ &= \int_{-\infty}^{\infty} e^{s(t-u)} \cdot \boldsymbol{h(u)} du \\ &= e^{st} \underbrace{\int_{-\infty}^{\infty} \boldsymbol{e^{-su} \cdot h(u) du}}_{\boldsymbol{H(s)}} \\ &= \boldsymbol{H(s)} e^{st}. \end{aligned}$$

Motivation

- **Continuous-time**:
  - We have output as $y(t) = \boldsymbol{H(s)}e^{st}$.
  - Output signal is scalar multiple of input signal.
  - Scalar $H(s)$ is a function of the parameter s.
  - Signals e^{st} are the **eigensignals** (**eigenfunctions**) for the LSI (LTI) system.
  - The scalars $H(s)$ are called the **eigenvalues** corresponding to the **eigensignals** e^{st}.

Motivation

- **Discrete-time**:
  - $$y[n] \quad = x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] \cdot \boldsymbol{h[n-m]}.$$
- Set input $x[n] = z^n$. We have output as

$$\begin{aligned} y[n] &= \sum_{m=-\infty}^{\infty} z^n \cdot \boldsymbol{h[n-m]} \\ &= \sum_{m=-\infty}^{\infty} z^{n-m} \cdot \boldsymbol{h[m]} \\ &= z^n \underbrace{\sum_{m=-\infty}^{\infty} \boldsymbol{z^{-m} \cdot h[m]}}_{\boldsymbol{H(z)}} \\ &= \boldsymbol{H(z)} z^n. \end{aligned}$$

Motivation

- **Discrete-time**:
  - We have output as $y[z] = \boldsymbol{H(z)}z^n$.
  - Output signal is scalar multiple of input signal.
  - Scalar $H(z)$ is a function of the parameter z.
  - Signals z^n are the **eigensignals** (**eigenfunctions**/**eigensequences**) for the LSI (LTI) system.
  - The scalars $H(z)$ are called the **eigenvalues** corresponding to the **eigensignals** z^n.

Even and Odd Signals

- **Even signals** $f(t)$, $f[n]$: if $f(-t) = f(t)$ for all real t and $f[-n] = f[n]$ for all integer n.
- **Odd signals** $f(t)$, $f[n]$: if $f(-t) = -f(t)$ for all real t and $f[-n] = -f[n]$ for all integer n.
- **Trigonometric signals**: cos and sin.

Continuous-time Periodic Signals

- **Continuous-time Periodic Signals**
  - $x(t)$ with a period T, $T > 0$: $x(t+T) = x(t)$ for all t.
    - e.g. $\cos(t)$ and $\sin(t)$.
  - period $T \Rightarrow$period nT for all integers $n > 0$.
  - **Scalar constant multiples**: still same periodic
  - **Addition of two or more**: still periodic with a least common period
  - Effectively, finite **linear combinations preserve periodic property**, albeit with a common period.
- **Trigonometric periodic signals**: cos and sin.
  - Retain sinusoidal properties on integration and differentiations.

Trigonometric Signals

- Consider the set $\mathcal{F}$ of signals:
  - $\cos(nt)$ for any integer n, $0 \leq n \leq N$, and
  - $\sin(nt)$ for any integer n, $1 \leq n \leq N$
  - defined for all real t and a fixed integer N.
- **Period**: $\dfrac{2\pi}{n}, n > 0$ and **constant:** for $n = 0$**.**
  - Hence, for each $f \in \mathcal{F}$**, we have period** 2π**.**

- **Orthogonality relationships in a period:**
  - For $f, g \in \mathcal{F}$, $\displaystyle\int_{-\pi}^{\pi} f(t)g(t) \cdot dt = \begin{cases} 0, & f \neq g \\ c, & f = g \end{cases}$ with

    $c = \begin{cases} 2\pi, & f = 1 \\ \pi, & \text{otherwise.} \end{cases}$
  - Meaning, each component cannot be obtained linearly from others as in next slide.

Orthogonal Set of Continuous-time Signals

- Let's synthesise a signal h as a **linear combination** of the set $\mathcal{F}$.
  - $h(t) := \sum_{i=1}^{K} \alpha_i f_i(t)$ for some real α_i and finite positive integer K with $f_i \in \mathcal{F}$ and $f_i \neq f_j$, $i \neq j$.
  - α_i are the linear co-efficients corresponding to the linear component f_i.
  - **Period:** 2π**.**
- **Are α_i unique?**
  - Can take $\int_{-\pi}^{\pi} h(t) f_i(t) \cdot dt = \int_{-\pi}^{\pi} \left(\sum_{i=1}^{K} \alpha_i f_i(t) f_i(t) \right) \cdot dt = \alpha_i c$ with $c = \begin{cases} 2\pi, & f_i = 1 \\ \pi, & \text{otherwise.} \end{cases}$
- **Size of the set** of such linear combinations?

Continuous-time Fourier Series

- Consider linear combinations in $\mathcal{F}$ when $N = \infty$.

$$\begin{aligned} h(t) &:= \frac{1}{2}A_0 + \sum_{n=1}^{\infty} A_n \cos nt + \sum_{n=1}^{\infty} B_n \sin nt \\ &= \frac{1}{2}A_0 + \sum_{n=1}^{\infty} \left(A_n \cos nt + B_n \sin nt\right). \end{aligned}$$

  - RHS is called **Fourier Series** for h.
  - A_0, A_m, B_m are called the **Fourier Series Coefficients**.
  - Is the series convergent?

Continuous-time Fourier Series

- How to find A_0, A_m, B_m?

$$\begin{aligned}
& \int_{-\pi}^{\pi} h(t)dt \\
= \quad & \frac{1}{2}A_0 \int_{-\pi}^{\pi} \mathbf{1} \cdot dt + \sum_{n=1}^{\infty} A_n \int_{-\pi}^{\pi} \cos nt \cdot dt + \sum_{n=1}^{\infty} B_n \int_{-\pi}^{\pi} \sin nt \cdot dt \\
& \qquad \text{(somehow!)} \\
= \quad & \boldsymbol{A_0 \pi} + 0 + 0 = A_0 \pi.
\end{aligned}$$

- i.e. $\boldsymbol{A_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} h(t)dt}$.

Continuous-time Fourier Series

- How to find A_0, A_m, B_m?

$$\int_{-\pi}^{\pi} h(t)\cos mt dt$$

$$= \frac{1}{2}A_0\int_{-\pi}^{\pi}\cos mt\cdot dt + \sum_{n=1}^{\infty}A_n\int_{-\pi}^{\pi}\cos nt\cos mt\cdot dt$$

$$+\sum_{n=1}^{\infty}B_n\int_{-\pi}^{\pi}\sin nt\cos mt\cdot dt$$

(somehow!)

$$= 0 + \boldsymbol{A_m\pi} + 0.$$

$$= \boldsymbol{A_m\pi}.$$

- i.e. $\boldsymbol{A_m = \frac{1}{\pi}\int_{-\pi}^{\pi} h(t)\cos mt dt}$.
- Similarly, $\boldsymbol{B_m = \frac{1}{\pi}\int_{-\pi}^{\pi} h(t)\sin mt dt}$.

Continuous-time Fourier Series

- **Complex form**:
  - $e^{j\theta} = \cos(\theta) + j\sin(\theta)$ for all real θ.
    - Effectively, we have

$$\begin{aligned}\cos(nt) &= \frac{e^{jnt} + e^{-jnt}}{2} \\ j\sin(nt) &= \frac{e^{jnt} - e^{-jnt}}{2}.\end{aligned}$$

$$\begin{aligned}h(t) &= \frac{1}{2}A_0 + \sum_{n=1}^{\infty}\left(A_n \cos nt + B_n \sin nt\right). \\ &= \frac{1}{2}A_0 + \sum_{n=1}^{\infty}\left(A_n \frac{e^{jnt} + e^{-jnt}}{2} - jB_n \frac{e^{jnt} - e^{-jnt}}{2}\right) \\ &= \frac{1}{2}A_0 + \sum_{n=1}^{\infty}\left(\frac{A_n - jB_n}{2}\right)e^{jnt} + \sum_{n=1}^{\infty}\left(\frac{A_n + jB_n}{2}\right)e^{-jnt} \\ \boldsymbol{h(t)} &= \sum_{n=-\infty}^{\infty} \boldsymbol{c_n e^{jnt}} \text{ where } c_0 = \frac{1}{2}A_0, c_n = \frac{A_n - jB_n}{2}, c_{-n} = \frac{A_n + jB_n}{2}, n > 0.\end{aligned}$$

Continuous-time Fourier Series

$$h(t) = \sum_{n=-\infty}^{\infty} c_n e^{jnt}$$

- **Complex form - orthogonality**:
  - $\int_{-\pi}^{\pi} e^{jmt} e^{jnt} \cdot dt = \int_{-\pi}^{\pi} e^{j(m+n)t} \cdot dt = \begin{cases} 2\pi, & m+n=0 \\ 0, & \text{otherwise.} \end{cases}$
- **Co-efficients**: $\boldsymbol{c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} h(t) e^{-jnt} \cdot dt}$

Continuous-time Fourier Series

- Fourier Series:

$$h(t) = \sum_{n=-\infty}^{\infty} c_n e^{jnt}$$

  - Fundamental Period: $T = 2\pi$.
- We can rescale time to in general to get any fundamental period T.

$$\boldsymbol{h(t) = \sum_{n=-\infty}^{\infty} c_n e^{-j\frac{2\pi nt}{T}}}.$$

- Fundamental frequencies $f_1 := \pm\frac{1}{T}$
- Fundamental angular frequencies $\omega_1 := \pm\frac{2\pi}{T}$
- Harmonic frequencies $f_n := \pm\frac{n}{T}$ for integers $n > 0$.
- Harmonic angular frequencies $\omega_n := \pm\frac{2\pi n}{T}$.

Continuous-time Fourier Series

- Fourier Series:

$$h(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi nt}{T}}.$$

  - Fundamental frequencies $f_1 := \pm\frac{1}{T}$
  - Fundamental angular frequencies $\omega_1 := \pm\frac{2\pi}{T}$
  - Harmonic frequencies $f_n := \pm\frac{n}{T}$ for integers $n > 0$.
  - Harmonic angular frequencies $\omega_n := \pm\frac{2\pi n}{T}$.
  - **Co-efficients**: $c_n = \frac{1}{T}\int_{-T/2}^{T/2} h(t)e^{-jn\omega_0 t} \cdot dt$

Continuous-time Fourier Series

- When does the Fourier Series converges in mean-squared, pointwise?
- **Jump discontinuity**
  - A signal $f(t)$ has a jump discontinuity at t_0 when $f(t_0^-) \neq f(t_0^+)$ both existing.
- **Piecewise continuous**
  - A signal $f(t)$ is piecewise continuous in interval (t_1, t_2) when f has a only finite number of jump discontinuities, no other discontinuities and both $f(t_1^+)$ and $f(t_2^-)$ existing.

Continuous-time Fourier Series

- **Finite energy in single period periodic signals**
  - A signal $\boldsymbol{h(t)}$ with finite energy in single period
    $$E_T = \int_{-T/2}^{T/2} |h(t)|^2 \cdot dt < \infty.$$
  - The truncation error signal is given as
    $$e_N(t) = h(t) - \sum_{n=-N}^{N} c_n e^{jn\omega_0 t}.$$
  - For N, error
    $$\int_{-T/2}^{T/2} |e_N(t)|^2 \, dt = \int_{-T/2}^{T/2} \left| h(t) - \sum_{n=-N}^{N} c_n e^{jn\omega_0 t} \right|^2 dt.$$
    - Minimise for minimum error $\Rightarrow c_n = \dfrac{1}{T} \displaystyle\int_{-T/2}^{T/2} h(t) e^{-jn\omega_0 t} \cdot dt$
  - $e_N(t)$ converges to 0 in mean-squared sense (energy in one time period). i.e. $\boldsymbol{\displaystyle\int_{-T/2}^{T/2} |e_\infty(t)|^2 \, dt = 0.}$
- How about any $h(t)$? How pointwise?

Continuous-time Fourier Series

- $\int_{-T/2}^{T/2} |e_\infty(t)|^2 \, dt = 0$ means if there is any different in representation the energy of it is zero.
- How about any $h(t)$? How pointwise?
- **Dirichlet conditions (1)**:
  - $h(t)$ is absolutely integrable in a period, $\int_{-T/2}^{T/2} |h(t)| \, dt < \infty$.
    - This results in the following:

$$
\begin{aligned}
|c_n| &= \left| \frac{1}{T} \int_{-T/2}^{T/2} h(t) e^{-jn\omega_0 t} \cdot dt \right| \\
&\leq \frac{1}{T} \int_{-T/2}^{T/2} \left| h(t) e^{-jn\omega_0 t} \right| \cdot dt \\
&\leq \frac{1}{T} \int_{-T/2}^{T/2} |h(t)| \cdot dt < \infty.
\end{aligned}
$$

  - Example not satisfying: $h(t) := \frac{1}{t}$, $0 < t < 1$ with period 1.

- **Dirichlet conditions (2)**:
  - $h(t)$ has **bounded variation** in any finite interval.
  - Meaning, $h(t)$ can take only a finite number of extremum values (minima, maxima) in a single period.
  - Example not satisfying: $h(t) := \sin \frac{1}{t}$, $0 < t \leq \frac{1}{2\pi}$ with period $\frac{1}{2\pi}$.

- **Dirichlet conditions (3)**:
  - $h(t)$ has only a **finite number** of **finite** jump discontinuities in a finite interval.
  - Example not satisfying:

$$h(t) := \begin{cases} 1, & 0 < t \leq 0.5 \\ 0.5 & 0.5 < t \leq 0.75 \\ \dots & \dots \end{cases}, \; 0 < t \leq 1 \text{ with period } 1.$$

Continuous-time Fourier Series

- **Dirichlet condition** implies:
  - Fourier series of $h(t)$ converges to $h(t)$ pointwise everywhere except at points of discontinuities.
  - at the points t_i of discontinuities, Fourier series of $h(t)$ converges to $\frac{1}{2}\left(h(t_i^-) + h(t_i^+)\right)$.

- We write the fourier series transformation as $h(t) \xleftrightarrow{\mathcal{FS}} c_n$

- **Fourier Series – Properties**:
  - **Linearity**: $h_1(t) \xleftrightarrow{\mathcal{FS}} c_{1,n}$ and $h_2(t) \xleftrightarrow{\mathcal{FS}} c_{2,n}$ implies $\alpha h_1(t) + \beta h_2(t) \xleftrightarrow{\mathcal{FS}} \alpha c_{1,n} + \beta c_{2,n}$.
  - **Shifting**: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies $h(t - t_0) \xleftrightarrow{\mathcal{FS}} e^{-jn\omega_1 t_0} c_n$.
  - **Time-reversal**: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies $h(-t) \xleftrightarrow{\mathcal{FS}} c_{-n}$.
  - **Time-scaling**: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies $h(at) \xleftrightarrow{\mathcal{FS}} c_n$.

- **Fourier Series – Properties**:
  - **Multiplication**: $h_1(t) \xleftrightarrow{\mathcal{FS}} c_{1,n}$ and $h_2(t) \xleftrightarrow{\mathcal{FS}} c_{2,n}$ implies $h_1(t)h_2(t) \xleftrightarrow{\mathcal{FS}} \sum_{k=-\infty}^{\infty} c_{1,k} \cdot c_{2,(n-k)}$.
  - **Conjugation**: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies $h^*(t) \xleftrightarrow{\mathcal{FS}} c^*_{-n}$.
  - **Real signal**: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies $c_n = c^*_{-n}$.
  - **Real and even signal**: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies c_n real and even.
  - **Real and odd signal**: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies c_n imaginary and odd.

- **Fourier Series – Properties**:
  - Parseval's Relation: $\frac{1}{T}\int_{-T/2}^{T/2} |h(t)| \cdot dt = \sum_{k=-\infty}^{\infty} |c_n|^2$.
  - Differentiation: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies $\frac{d}{dt}h(t) \xleftrightarrow{\mathcal{FS}} jn\omega_1 c_n$.
  - Integration: $h(t) \xleftrightarrow{\mathcal{FS}} c_n$ implies $\int_{-\infty}^{t} h(t) \xleftrightarrow{\mathcal{FS}} \frac{1}{jn\omega_1} c_n$.
    (integration result has to be finite valued and periodic)